Task 11: Docker Image and Container Lifecycle

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1. Objective

Containerize the Spring Boot application, build the Docker image, run a named container, and verify the application.

2. Dockerfile

```
FROM eclipse-temurin:21-jre

WORKDIR /app

COPY target/delivery-promise-calculator-0.0.1-SNAPSHOT.jar app.jar

EXPOSE 8765

ENTRYPOINT ["java", "-jar", "app.jar"]
```

3. Build and Run

```
docker build -t delivery-promise-calculator:1.0 .
docker rm delivery-promise-app
docker run -d --name delivery-promise-app -p 8766:8765 delivery-promise-calculator:1.0
docker ps
```

4. Port Mapping

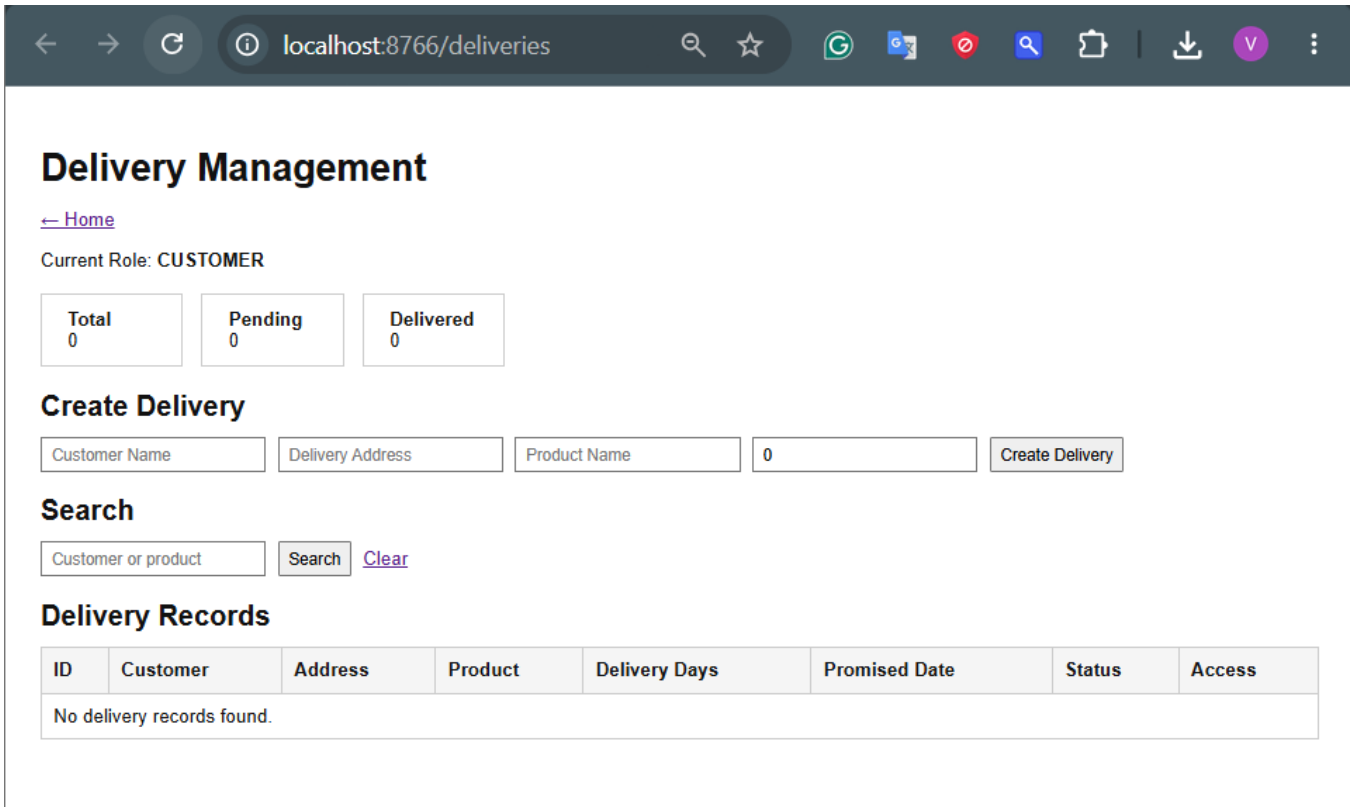
The application listens on 8765 inside the container. Host port 8766 was selected because the Jenkins-deployed application already occupied host port 8765.

5. Verification URL

```
http://localhost:8766/deliveries
```

6. Evidence

Evidence: Docker deployment accessed through the browser



7. Result

The Docker image and container were successfully built/run and the Delivery Management page loaded from the container.

Task 11 — Actual Project Screenshots

Screenshots captured during the implementation are included below as evidence. These are the actual project screenshots, not recreated diagrams.

Figure 1. Docker container/image verification from the command line.

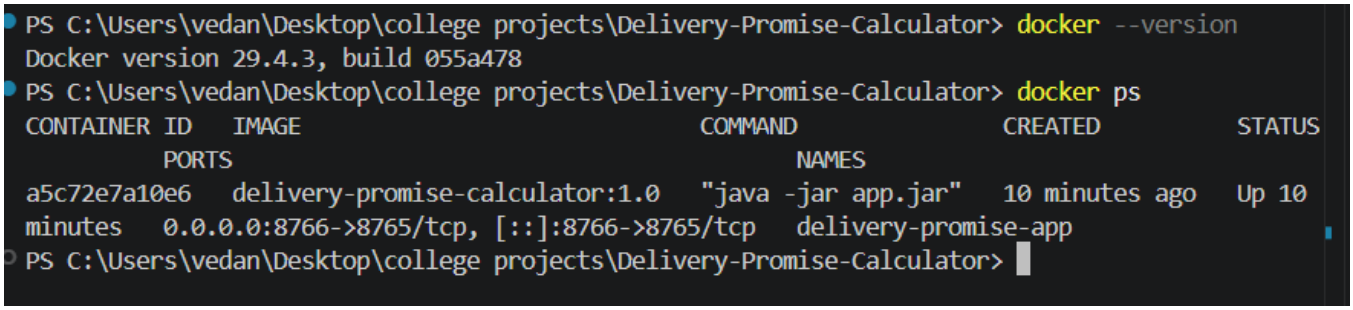


Figure 2. Delivery Management page served from the Docker deployment on localhost:8766.

